From Research Wheel Length = 0.3m

Material SHN model uses Aluminium ≈ 1" or 25.4mm or 0.0254m thick

SF=4 $\delta_{max} = 0.1mm = 1 \cdot 10^{-4} m$

$\sigma_{max} = 5.5 \cdot 10^7 \, N/m^2$

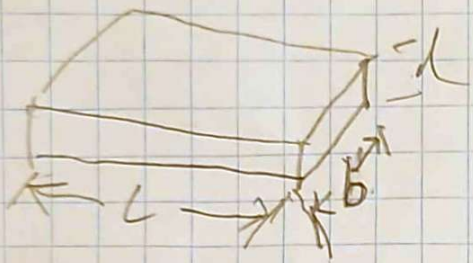
Width = Nut Flange $+2.6^{+0.00}_{-0.00}$ mm Wb = Calc

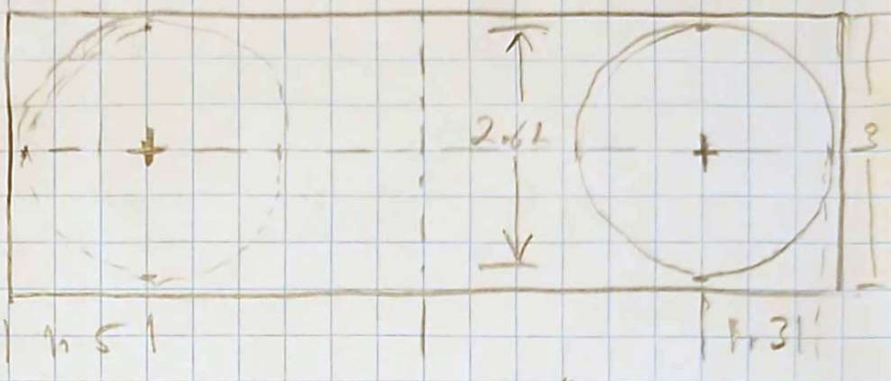
Static 76.2mm or 0.0762 m

TAKING "FREELOADER" AS EXAMPLE

WIDTH = 3"
FLANGE WIDTH 2-5/8 OVG = 3/16 - 5/16 ≈ 4.3/4mm

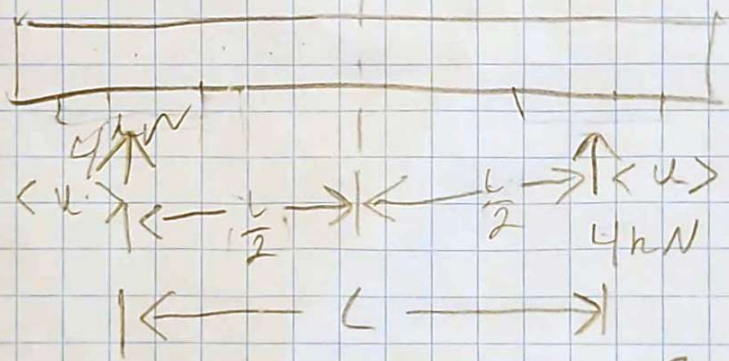
LENGTH = 9"
SPACING NUT / FLANGE = 6"





LENGTH SIDE OVG = 0.19" / SIDE ≈ EQUAL TO WIDTH ≈ 10mm OVG OF WIDTH & LENGTH PAST FLANGES

FBD



$Z = \frac{6d^2}{3}$

$I = \frac{6d^3}{12}$

$MAX\ \sigma = \frac{-WL}{4Z}$; $MAX\ \delta = \frac{-WL^3}{48EI}$

KNOWN $\sigma_y = 276 \cdot 10^6 \ N \cdot m^{-2}$ σ_y & G from MATERIAL

6061 T6 $L = 0.3\ m$ $\delta_{max} = 1 \cdot 10^{-4}\ m$

Load = 2 kN $E = 68.9 \cdot 10^9 \ N \cdot m^{-2}$ $Z = \frac{bd^2}{3}$

SF = 4

$W = Load \cdot SF = 8 \cdot 10^3 \ N$

$b(est) = L/3 = 0.1\ m$ $I = \frac{bd^3}{12}$

UNKNOWN d, Z, I

EQNS STRESS: $\sigma_y = \frac{WL}{4Z}$ DEFLECTION $\delta_{max} = \frac{WL^3}{48EI}$

SOLVE FOR Z $Z = \frac{WL}{4\sigma_{max}}$

SOLVE FOR I $I = \frac{WL^3}{48E\delta_{max}}$

Solve For d $\frac{bd^2}{3} = \frac{WL}{4\sigma_y}$ $d = \left(\frac{3WL}{4b\sigma_y}\right)^{1/2}$

Solve For d $\frac{bd^3}{12} = \frac{WL^3}{48E\delta_{max}}$ $d = \left(\frac{12WL^3}{48bE\delta_{max}}\right)^{1/3}$

Add Numerical

d_σ

$$d_\sigma = \left(\frac{3(8 \cdot 10^3 N)\,0.3\,m}{4 \cdot 0.1\,m \cdot 276 \cdot 10^6 N \cdot m^{-2}}\right)^{1/2}$$

$d_\sigma = 0.0081\ m = 8.1\ mm$

d_δ

$$d_\delta = \left(\frac{12 \cdot (8 \cdot 10^3 N)(0.3\,m)^3}{48(68.9 \cdot 10^9 N \cdot m^{-2})(0.1\,m)(0.0001\,m)}\right)^{1/3}$$

$d_\delta = 0.0428\ m = 42.8\ mm$

$d_\sigma < d_\delta$, d_δ will be used

42.8 mm -or- 1.684 in

AVAILABLE MAT'L McMASTER 5679N148

$1\frac{3}{4}$" THK x 4" WIDE x 12" L

VS CALC 1.68" 3.937" 11.8